Requirements Gathering

CineMatch – Hybrid Movie Recommender System

1. Introduction

The purpose of the Requirements Gathering phase is to identify and understand the needs of the intended users and define the essential requirements of the CineMatch system.

CineMatch aims to help users discover movies that match their interests by combining Content-Based Filtering and User-Based Collaborative Filtering into a single recommendation system.

The gathered requirements will guide the system design, model development, application implementation, and final testing.

2. Requirements Gathering Approach

The project requirements were identified through the following activities:

2.1 Data Source Analysis

Examining the available datasets and their structure to identify the information required for building the recommendation system. The main data sources include:

- **MovieLens:** Movie information and user ratings.
- **TMDB:** Movie descriptions, genres, keywords, cast, and directors.
- **IMDb:** Movie ratings and additional movie information.

2.2 Existing Recommendation Systems Analysis

Reviewing common recommendation approaches and studying how movie platforms help users discover relevant content. The analysis focused on:

- Content-Based Recommendation.
- Collaborative Filtering.
- Hybrid Recommendation Systems.
- Personalized movie discovery.

2.3 User Needs Identification

Identifying the main expectations of different user types, including casual viewers and users interested in discovering movies with specific themes or genres.

2.4 Technical Requirements Analysis

Identifying the technical requirements needed to process movie data, build recommendation models, and provide an interactive application.

3. Stakeholder Identification

The following stakeholders were identified:

Stakeholder	Needs and Expectations
Casual Movie Viewer	Quickly discover movies based on personal interests.
Movie Enthusiast	Explore movies from different genres, directors, and themes.
DEPI Evaluators	Review the project methodology, implementation, and results.
Development Team	Build a reliable, maintainable, and efficient recommendation system.

4. Identified User Requirements

Based on the identified stakeholder needs, the following user requirements were established:

- Users should be able to search for movies.
- Users should be able to rate selected movies.
- Users should receive personalized movie recommendations.
- Users should be able to discover movies from different genres.
- Users should be able to view basic movie information and ratings.
- New users should be able to receive recommendations even when they have limited rating history.
- The application should provide a simple and user-friendly interface.

5. Functional Requirements

The system must provide the following functionalities:

ID	Requirement	Description
FR-01	Data Collection	Collect movie information and ratings from MovieLens, TMDB, and IMDb.
FR-02	Data Preprocessing	Clean and prepare movie metadata and rating datasets.
FR-03	Content-Based Filtering	Recommend movies based on movie descriptions, genres, cast, directors, and keywords.
FR-04	Collaborative Filtering	Identify users with similar movie preferences based on their ratings.
FR-05	Hybrid Recommendation	Combine Content-Based and Collaborative Filtering results.
FR-06	Movie Search	Allow users to search for movies by title.
FR-07	Movie Rating	Allow users to submit ratings for selected movies.
FR-08	Personalized Recommendations	Generate a ranked list of recommended movies.
FR-09	Cold-Start Handling	Provide recommendations when collaborative information is unavailable by using Content-Based Filtering.
FR-10	Recommendation Display	Display recommended movies with relevant information and ratings.

6. Non-Functional Requirements

The system should satisfy the following quality requirements:

ID	Category	Requirement
NFR-01	Performance	Generate recommendations within an acceptable response time, targeting less than 2 seconds.
NFR-02	Usability	Provide a simple and intuitive user interface.
NFR-03	Reliability	Handle missing data and invalid user inputs without application crashes.
NFR-04	Efficiency	Optimize memory usage when loading and processing recommendation models.
NFR-05	Maintainability	Organize the code into separate modules for data processing, recommendation models, and application logic.
NFR-06	Portability	Support execution in standard Python environments.

7. Technical Requirements

The project requires the following technologies and tools:

Category	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Natural Language Processing	TF-IDF, NLTK, VADER
Similarity Calculation	Cosine Similarity
Sparse Matrix Operations	SciPy
Web Application	Streamlit
Data Sources	MovieLens, TMDB, IMDb
Version Control	Git and GitHub
Deployment	Streamlit Community Cloud

8. System Constraints

The following constraints were identified during requirements analysis:

- The recommendation system depends on the availability and quality of movie metadata and user ratings.
- Collaborative Filtering performance may be affected by the sparsity of the user-item rating matrix.
- New users may have insufficient rating history to generate collaborative recommendations.
- Large similarity matrices may require memory optimization.
- External movie information depends on the available datasets or API access.

9. Assumptions

The project is based on the following assumptions:

- MovieLens, TMDB, and IMDb data will be available for development.
- Users can provide movie ratings through the application.
- Movie titles and identifiers can be matched across the required datasets.
- The recommendation models will be trained and tested using the available datasets.
- The application will run in a Python-compatible environment.

10. Expected Outcomes

After gathering and analyzing the requirements, the project is expected to deliver:

1. A cleaned and integrated movie dataset.
2. A Content-Based Recommendation Model.
3. A User-Based Collaborative Filtering Model.
4. A Hybrid Recommendation Engine.
5. An interactive Streamlit application.
6. Personalized Top-10 movie recommendations.
7. A system that can handle new users through Content-Based recommendations.

11. Conclusion

The Requirements Gathering phase established the functional, non-functional, and technical needs of the CineMatch project.

These requirements provide a clear foundation for the system design and development phases, ensuring that the final application addresses user needs while meeting the project's technical and academic objectives.